DelaySlayer

AI-powered flight delay complaint handling, built on EU 261/2004

Problem Statement

Flight delay complaints are one of the highest-volume, most legally structured complaint types airlines receive. Yet most airlines still handle them manually: a human reads the complaint, checks the flight data, looks up EU 261/2004 rules, and writes a response. This process is slow, inconsistent, and expensive.

At the same time, passengers rarely understand their rights and often either abandon valid claims or submit incomplete ones. During a disruption event, complaint volume spikes exactly when airline staff are most stretched. The result is backlogs, frustrated passengers, and preventable escalations.

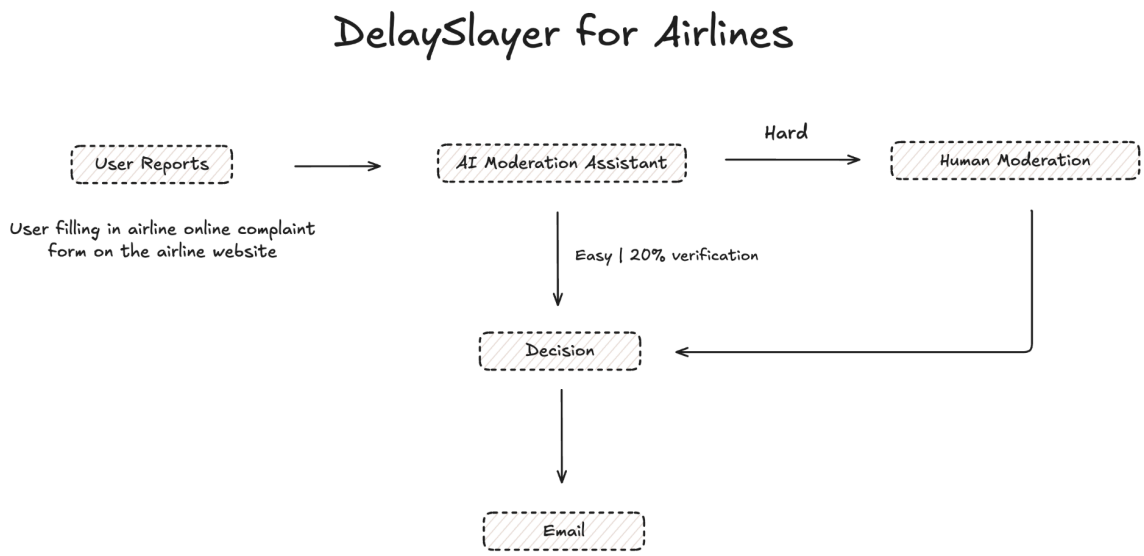
EU 261/2004 is actually well-suited for automation. The eligibility rules are explicit, the compensation tiers are fixed, and the data needed to verify a claim is available via API. The gap is not knowledge, it is execution.

What We're Building

Two products, one legal engine.

- **DelaySlayer for Airlines** is a B2B AI moderation assistant sold to airlines operating at Schiphol. It processes incoming delay-related complaints end to end and either resolves them automatically or hands them to a human moderator with a fully prepared recommendation.
- **DelaySlayer for Passengers** is a B2C chatbot deployed on airport apps or websites. When a flight is delayed, passengers can enter their flight details and instantly find out whether they are eligible for compensation, how much they can claim, and where to submit their complaint.

Product 1: DelaySlayer for Airlines



Workflow

Step 1: Automated eligibility check (Agent 1)

When a complaint arrives, the first agent pulls real-time flight data via API to verify two hard eligibility conditions under EU 261/2004:

1. Does the flight originate from or arrive in an EU member state (including the UK under retained law)?
2. Was the passenger delayed by 3 hours or more at final destination?

If either condition is not met, the case is automatically closed. The system generates a decision with clear reasoning and sends a response email to the passenger explaining why their claim does not qualify. No human involvement needed.

If both conditions are met, the case moves forward.

Step 2: Complexity triage and compensation decision (Agent 2)

The second agent assesses the case and classifies it as easy or hard.

For easy cases, the agent makes a full determination autonomously: it applies EU 261/2004 compensation thresholds based on flight distance and delay duration, checks for extraordinary circumstances exemptions (e.g. severe weather, air traffic control strikes), and produces a compensation recommendation or a reasoned rejection.

For hard cases (e.g. connecting flights, partial delays, contested extraordinary circumstances), the case is routed to a human moderator. The moderator receives the full case file, the AI's preliminary assessment, and a one-click option to adopt the recommendation or override it.

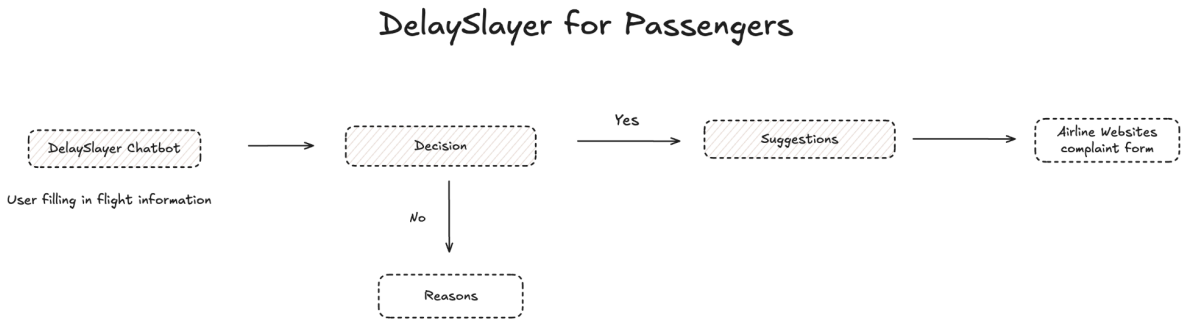
Step 3: Fraud check

Before any decision is finalised, a fraud detection layer screens the claim for patterns associated with abuse: duplicate submissions, mismatched passenger details, bulk claim patterns, and inconsistencies between stated and verified flight data.

Step 4: Output

The system generates a structured decision with full reasoning, a draft response email to the passenger, and an audit log. All outputs are traceable to the model version and prompt used.

Product 2: DelaySlayer for Passengers



A chatbot available on airport apps and websites. When a flight is delayed, passengers enter their flight number and delay duration. The chatbot:

1. Checks EU 261/2004 eligibility in real time
2. Tells the passenger clearly whether they are eligible and why
3. If eligible, calculates the expected compensation amount
4. Directs the passenger to the correct complaint submission page for their airline, with a step-by-step guide on what to include

The goal is to close the information gap for passengers and increase the quality and legitimacy of complaints that reach airlines, which in turn makes the airline-side moderation more efficient.

Training Data

Since real labelled complaint data is limited and sensitive, we generate synthetic training cases using AI. These include a balanced mix of qualifying and non-qualifying complaints across different flight types, delay durations, routes, and passenger scenarios. This gives both products a robust starting point without relying on proprietary airline data.

Shadow Mode Experiment Design

Before going live, both products will run in shadow mode alongside existing processes.

For the airline product, the AI runs in parallel with human complaint handlers for 4 weeks. Its decisions are not shown to passengers and trigger no enforcement actions. Human moderators continue handling complaints as usual. At the end of each day, AI decisions are compared against human decisions to measure agreement rate, precision, and abstain rate.

Sampling ramp-up: Start at 10% of eligible delay complaints on day one. If guardrails are green after 3 days, increase to 30%. After 2 weeks, if precision holds, increase to 50%.

Success criteria: The model must achieve at least 90% agreement with human moderator decisions on cases where it does not abstain. If a case type consistently falls below 80%, it is removed from the automated scope and kept in shadow-only mode for further iteration.

Key metrics tracked

Precision on non-abstain decisions, abstain rate by case type, fraud detection accuracy, false positive rate on ineligible claims, average handling time reduction, and cost per resolved case.

For the passenger chatbot, shadow mode means running the eligibility check logic against a set of historical delay events with known outcomes before public launch, validating that compensation calculations and routing links are accurate across airlines and routes.

Expected Impact

For airlines: Manual complaint handling typically costs €15 to €30 per case when you account for agent time, tooling, and correspondence. At Schiphol alone, major carriers handle tens of thousands of delay complaints per year. Automating even 40% of straightforward cases reduces that cost significantly, and more importantly, it frees up complaints teams to focus on the genuinely complex cases that require human judgment. During disruption events, when

complaint volume can spike tenfold overnight, the AI handles the surge without creating a backlog.

	Manual Handling	**DelaySlayer**
Cost per case	€15 to €30	€0.02 to €0.05
Avg handling time per case	20 to 40 min	~10 seconds
Total cost per 10,000 cases	€150,000 to €300,000	€200 to €500
Total handling time per 10,000 cases	3,300 to 6,700 hours	~28 hours
Available 24/7	No	Yes

For airports: Schiphol and other European airports benefit from a passenger-facing product that reduces the chaos and confusion that accompanies large-scale delays. Fewer passengers crowding information desks asking about their rights means airport staff can focus on operational tasks. It also positions the airport as a passenger-first environment, which matters for reputation and commercial partnerships.

For passengers: Knowing your rights immediately after a delay changes the experience entirely. Instead of searching through confusing legal text or waiting days for an airline response, a passenger can open the app, enter their flight number, and know within seconds whether they have a valid claim and exactly what to do next. This reduces anxiety during an already stressful situation and ensures passengers do not abandon legitimate claims simply because the process feels too complicated.

For cabin crew and ground staff: A large part of the stress during delay events falls on frontline staff who are asked questions they cannot fully answer about compensation rights. When passengers already have access to clear, accurate information through the chatbot, those conversations become shorter and less confrontational. Crew can focus on what they are trained for rather than improvising legal explanations under pressure.

Technical Details

Flight data: Real-time and historical departure and arrival data is retrieved via flight status APIs (e.g. FlightAware, AviationStack, or airline-specific APIs). This is the ground truth for eligibility verification.

AI layer: The core reasoning engine uses a large language model prompted with the full text of EU 261/2004, structured flight data, and the complaint content. It outputs a structured JSON decision including eligibility verdict, compensation tier, reasoning, exemption assessment, and a draft passenger response.

Agent architecture: Two agents run in sequence. Agent 1 handles deterministic eligibility checks (route and delay threshold). Agent 2 handles legal reasoning and case complexity

classification. Hard cases are escalated with the AI's full reasoning attached so human moderators are not starting from scratch.

Fraud detection: A rule-based and ML-assisted layer checks for duplicate claims, mismatched booking references, coordinated submission patterns, and statistical anomalies in claimed delay durations versus verified data.

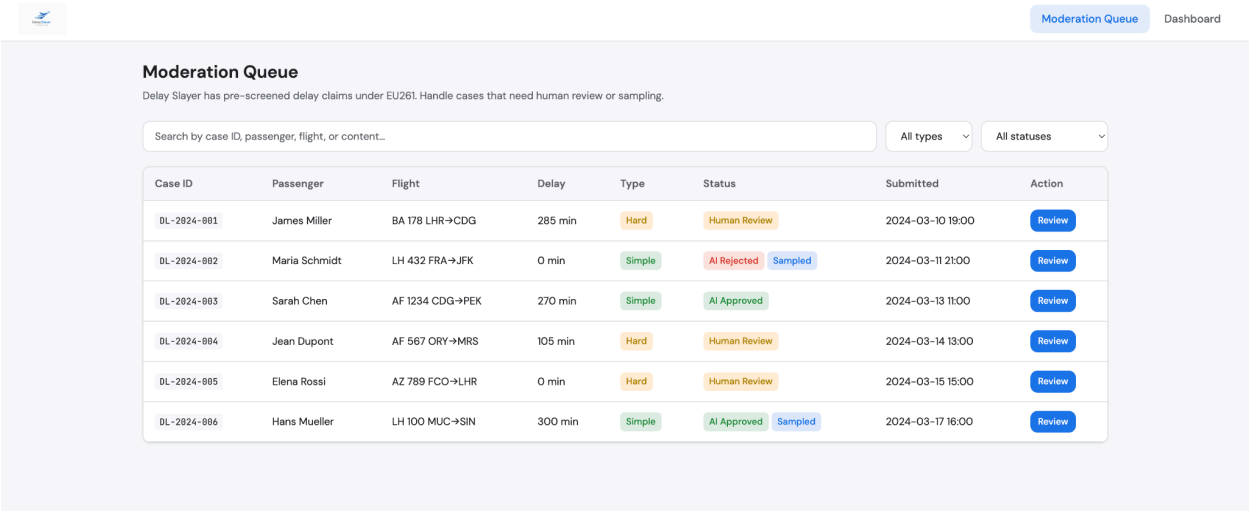
Synthetic training data: AI-generated complaint scenarios covering the full range of EU 261/2004 cases, including edge cases like missed connections, partial delays, and extraordinary circumstances, are used to fine-tune and evaluate the model before deployment.

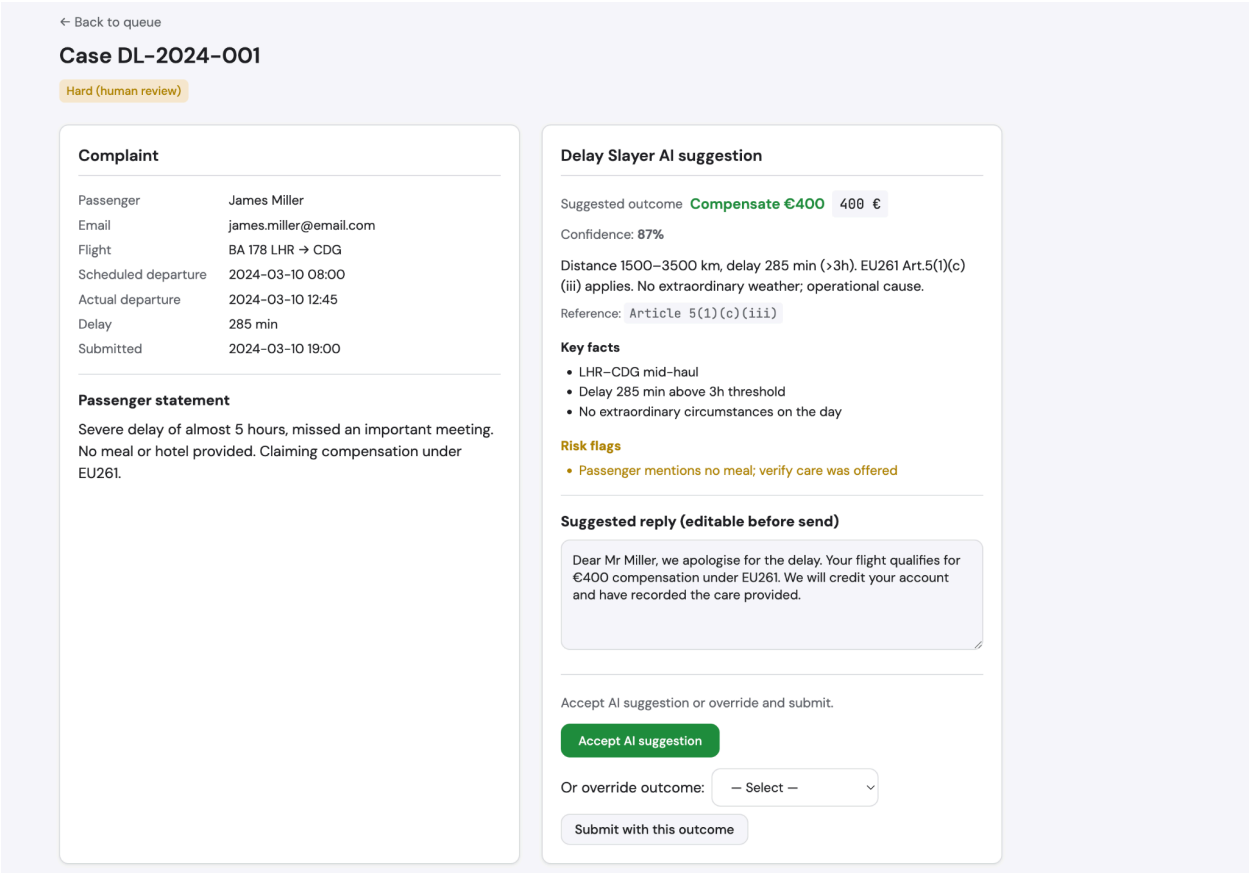
Auditability: Every decision is logged with its prompt version, model version, input data snapshot, and output. This ensures full traceability for regulatory review and internal auditing.

Infrastructure: Outputs are stored in a structured database. The passenger chatbot is deployable as an embeddable widget or standalone web app, compatible with existing airport and airline digital platforms.

Live Demo

To validate the concept, we built a working prototype of the airline-facing moderation platform. The demo includes a full moderation queue where cases are routed as simple or hard, an AI suggestion panel showing EU 261/2004 reasoning, suggested compensation outcome, key facts, risk flags, and a draft reply that human moderators can accept in one click. The dashboard tracks AI accuracy against human coder decisions, daily processing volume, cost per case, appeal rate, appeal success rate, and passenger satisfaction in real time.





The prototype runs on synthetic complaint data generated from live AviationStack flight records, giving a realistic picture of what the production system would handle. It is deployable as a web app and designed to sit alongside any existing airline complaint management workflow.

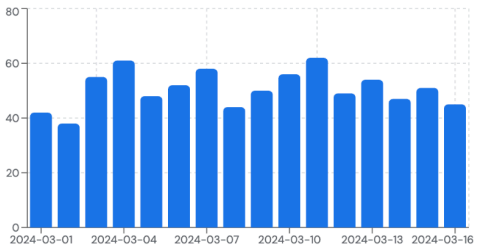
Delay Slayer dashboard

AI accuracy, human consistency, volume, cost, and passenger metrics

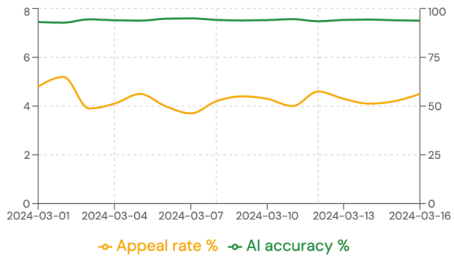
AI Accuracy	Human Coder Consistency	AI Processed Volume	Human Review Volume	Total Cost (€)
94.2%	**91.8%**	**1,247**	**312**	**28,450**
Agreement with human coding (sampled cases)	Inter-coder agreement on sampled cases	Cases auto-processed by AI this period	Hard cases + 20% sample for verify	Cost per case €18.2

Appeal Rate	Satisfaction	Appeal Success Rate
4.3%	**4.2**	**12.5%**
Share of decisions appealed by passengers	Out of 5, post-response survey	Share of appeals that overturn decision

Daily volume



Appeal rate vs AI accuracy (trend)



Period summary

Simple cases (AI auto): **935**

Hard cases (human): **312**

Sampled for verify (20% of simple): **187**